

Lab-3 Ex2 - Quadratic Equations (II)

(PreLab*)

This is a PreLab exercise. Please do the PreLabs after lecture and before you come to lab!

In Lab-2, you have worked on calculating the **discriminant** of a quadratic equation with **coefficients** a , b and c is defined as:

$$\Delta = b^2 - 4ac$$

Below you will find a slightly extended C program that calculates not only the **discriminant** of a quadratic equation with **coefficients** a , b , c but also the **number of real roots** of this quadratic equation:

```
#include <stdio.h>

int main(void) {
    double a,b,c,D;

    printf("a = ");
    scanf("%lf", &a);
    printf("b = ");
    scanf("%lf", &b);
    printf("c = ");
    scanf("%lf", &c);

    D = b * b - (4 * a * c);
    printf("discriminant is %.3f\n", D);
    printf("The quadratic equation has ");
    if (D > 0)
        printf("two real roots");
    else
        if (D==0)
            printf("one double real root");
        else {
            printf("no real roots");
        }
    printf(".\n");

    return 0;
}
```

As a warm-up, please type in the program above accurately so that it will correctly calculate the discriminant.

Optional improvement: the spacing, called indentation, in the if-else blocks of the example is not in good style. Can you fix it so that the code can be more readable?